



JAIDEV EDUCATION SOCIETY'S
**J D COLLEGE OF ENGINEERING AND
 MANAGEMENT**
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(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU & RTMNU

B. Tech Summer / Winter - 2025
(ESE / Make-up / Supplementary) Examination

Branch: CSE/IT/AI/CSE-DS/CE/EE/ME/ETC

Sem:-V

Subject: VEDIC MATHEMATICS

Subject Code: CS/IT/AI/DS/CE/EE/ME/ET50004

Marks: 100

Date of Exam: 20/11/2025

Time: 3 Hrs.

Instructions:

1. Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.
2. All the questions carry marks as indicated.
3. Assume suitable data whenever necessary.
4. Illustrate your answer whenever necessary with the help of neat sketches.
5. Use of Programmable Calculator is prohibited.
6. Don't use red pen for writing the answers.

COs: Students should be able to: (Listed Below CO & Mention it on question)

CO1: Describe concept of:

Demonstrate a comprehensive understanding of the history, principles, and significance of Vedic Mathematics. Recognize and explain the basic principles and concepts of Vedic Mathematics, including the 16 Sutras and their applications.

CO2: Illustrate the concept of:

Perform addition, subtraction, multiplication, and division mentally using Vedic Sutras with speed and accuracy. Apply mental calculation techniques to solve arithmetic problems efficiently.

CO3: Apply the knowledge of:

Apply advanced Vedic Mathematics techniques for squaring, finding square roots, cube roots, and factorization. Solve algebraic equations and manipulate algebraic expressions using Vedic methods.

CO4: Evaluate the problems:

Apply Vedic Mathematics techniques effectively to solve problems encountered in competitive exams such as aptitude tests, entrance exams, and Olympiads. Develop strategies to improve speed and accuracy in solving mathematical problems.

CO5: Analyze the problems and results:

Apply Vedic Mathematics principles to solve real-life problems in various fields such as finance, engineering, and science. Analyze and solve complex problems using Vedic Mathematics techniques, fostering critical thinking and problem-solving skills.

CO6: Create the methods or model:

Explore advanced topics in Vedic Mathematics, such as continued fractions and exponential series. Engage in research activities to explore modern extensions and applications of Vedic Mathematics in diverse areas.

SECTION A

Q. No.	All Questions are compulsory. Answer in One/Two/Sentence.	Level / CO Mapped	Marks
Q.1	Tell the Ekadhikena Purvena sutra and use it to find the square of 75.	L1/CO1	2

Q.2	Outline the Nikhilam Navatashcaramam Dashatah sutra .	L2/CO2	2
Q.3	Explain the Urdhva-tiryagbhyam method for multiplication and demonstrate it with 54×65 .	L2/CO2	2
Q.4	Define the Paravartya Yojayet sutra and use it to divide $5x^2 + 11x + 2$ by $x + 2$.	L1/CO1	2
Q.5	Outline the Shunyam Saamyasamuccaye sutra and solve the equation $6x + 10 = 6x + 20$.	L2/CO2	2
Q.6	Interpret the Anurupye Shunyamanyat sutra and use it to solve $\frac{5x}{4} = \frac{10x}{8}$.	L2/CO2	2
Q.7	List the key points of the Sankalana-vyavakalanabhyam sutra and apply it to solve $x + y = 35$ and $x - y = 7$.	L1/CO1	2
Q.8	Tell the Yavadunam sutra and use it to find the cube of 16.	L1/CO1	2
Q.9	Compare the Yavadunam sutra for squares and cubes.	L2/CO2	2
Q.10	Show how the Ekadhikena Purvena sutra can be used for decimal multiplication (e.g., 1.5×1.5).	L1/CO1	2
Q.11	Demonstrate the Nikhilam method for division (e.g., $112 \div 98$).	L2/CO2	2
Q.12	Summarize the Paravartya Yojayet method for linear divisors.	L2/CO2	2

SECTION B

Q. No.	All Questions are compulsory.	Level / CO Mapped	Marks
Q.13	Apply the Vedic mathematics sutra to multiply 4 by 9.	L3/CO3	4
Q.14	Use Nikhilam to multiply 997×994 . Explain the compensation process.	L3/CO3	4
Q.15	Demonstrate Urdhva-tiryagbhyam for 678×459 with complete working.	L3/CO3	4
Q.16	Analyze at least one Vedic mathematics sutra .	L4/CO4	4
Q.17	Compare the efficiency of Nikhilam for multiplication versus division of numbers near bases.	L4/CO4	4
Q.18	Differentiate between the application of Paravartya Yojayet for exact division versus division with remainder.	L4/CO4	4

SECTION C

Q. No.	Solve any FOUR Question.	Level / CO Mapped	Marks
Q.19	Critique the statement "Vedic Mathematics is only for simple calculations" by evaluating its application to compute 9999^2 using appropriate sutra.	L5/CO5	7
Q.20	Deduce the mathematical foundation of Anurupye Shunyamanyat sutra by solving $(2x+3)/(3x+4) = (4x+6)/(6x+8)$.	L5/CO5	7
Q.21	Estimate the computational savings achieved by using Urdhva-tiryagbhyam for 345×678 compared to conventional method.	L5/CO5	7
Q.22	Prioritize the following calculations based on their suitability for Vedic methods: (i) 125^2 (ii) 997×1003 (iii) $1/23$ (iv) $3x + 5y = 16$, $7x + 11y = 34$.	L5/CO5	7

Q.23	Prove the validity of Paravartya Yojayet sutra by dividing $2x^3 - 7x^2 + 5x - 3$ by $(x - 3)$ and verifying the result.	L5/CO5	7
Q.24	Rule on the effectiveness of Vedic Mathematics for engineering applications by assessing its performance in calculating 89×91 using Nikhilam sutra.	L5/CO5	7

SECTION D

Q. No.	Solve any TWO Question.	Bloom's Level & CO	Marks
Q.25	Explain at least two Vedic sutra in detail with example..	L5/CO5	12
Q.26	Discuss the use of Vedic mathematics in real life with suitable example.	L6/CO6	12
Q.27	Discuss history of Vedic Mathematics with suitable example.	L6/CO6	12